

PFE Project

Data Collection Brainstorming & Architecture

Sentiment Analysis for Financial Market Prediction

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1. Sector Selection Decision

After evaluating a multi-sector approach (5 sectors, 1 stock each), the decision was made to focus on a single sector. This improves academic depth, data coherence, and modeling quality compared to spreading thin across multiple sectors.

Sector Comparison Matrix

Sector	Sentiment-Price Link	Data Availability	Modeling Simplicity
Technology	★★★★★	★★★★★	★★★★
Financial/Banking	★★★★	★★★	★★★
Energy	★★★	★★★	★★
Healthcare/Pharma	★★★★	★★	★★
Crypto (Bitcoin)	★★★★★	★★★★★	★★

Priority ranking used for selection: 1st — Sentiment-price relationship is strong and proven | 2nd — Data availability | 3rd — Simplicity for modeling | 4th — Thesis differentiation

Selected sector: Technology — strongest academic precedent, best data availability, and proven sentiment-price correlation.

2. Stock Selection — Tesla (TSLA)

Within the Technology sector, Tesla (TSLA) was selected as the target stock based on the following justified criteria:

- **The Musk Effect:** Elon Musk's public statements cause measurable, near-immediate price movements in TSLA. This phenomenon is documented in published academic literature, directly strengthening the project's literature review connection.
- **Academic Precedent:** Multiple published research papers use TSLA specifically as their case study, providing strong methodological grounding and comparability.
- **Social Media Presence:** TSLA has extremely high coverage on Reddit (r/wallstreetbets, r/stocks, r/teslamotors) and in financial news, ensuring rich text data volume.
- **Clean Sentiment Dynamics:** Sentiment swings are large and clean enough to label BUY/SELL/HOLD signals clearly, which is critical for the target prediction variable.
- **vs NVIDIA (NVDA):** NVDA was considered but rejected — while strong currently due to the AI narrative, older labeled datasets from 2020–2022 are harder to find, weakening historical analysis.

3. Data Sources & Tools

Source	Tool	Cost	Coverage
TSLA Tweets	Kaggle / Hugging Face dataset (pre-collected)	Free	2019–2022

Source	Tool	Cost	Coverage
Reddit Posts	PRAW API	Free	Unlimited history
Financial News	GDELT Project / NewsAPI / Finnhub	Free	2017–present
OHLC Price Data	yfinance (Python library)	Free	Full history

4. The Elon Musk Tweet Problem

■ **Twitter/X closed free API access in 2023. Real-time tweet collection is no longer possible for free. Pre-collected datasets are the only practical solution for a PFE.**

Three realistic options were evaluated:

Option A — Pre-collected datasets (RECOMMENDED)

Use existing Kaggle or Hugging Face datasets containing TSLA-related tweets from 2019–2022. Elon's tweets are still present in the data, just sourced historically. Most practical for PFE timeline.

Option B — Reddit via PRAW

Still free and fully accessible. Reddit arguably provides richer, longer text than tweets. Subreddits like r/wallstreetbets offer deep financial sentiment signals.

Option C — News via GDELT/NewsAPI

GDELT is free, open, and covers financial news going back years. Good complement to social media sources.

Final Decision: Combine all three options — pre-collected tweet dataset (A) + Reddit via PRAW (B) + News via GDELT/NewsAPI (C). This ensures Elon's tweets remain in the dataset while maintaining full collection feasibility.

5. Data Collection Plan

Time Window: January 2020 — December 2023

This four-year window captures four distinct market regimes, making the model more robust and the thesis analysis more comprehensive:

- 2020 — COVID crash and recovery (extreme sentiment volatility)
- 2021 — TSLA all-time peak (euphoric sentiment period)
- 2022 — Major TSLA correction (negative sentiment dominance)
- 2023 — Recovery phase (mixed/stabilizing sentiment)

6. Label Definition Strategy (Trading Signals)

The target variable is BUY / SELL / HOLD. A precise, reproducible rule is required. The recommended academic approach:

$\text{Return} = (\text{next_day_close} - \text{today_open}) / \text{today_open}$ If $\text{Return} > +X\% \rightarrow \text{Label} = \text{BUY}$ If $\text{Return} < -X\% \rightarrow \text{Label} = \text{SELL}$ Otherwise $\rightarrow \text{Label} = \text{HOLD}$ Threshold X : typically 0.5% to 2% (design choice)

■ **Class Imbalance Risk:** TSLA will not move more than 2% every day. HOLD will dominate the dataset. Mitigation strategies required: oversampling (SMOTE), class-weighted loss functions, or threshold tuning.

7. Two-Layer Schema Architecture

Forcing one fixed schema on all data sources at collection time loses valuable source-specific information. A two-layer architecture is the correct engineering approach, following standard medallion/data lake principles:

Layer 1 — Raw (Source-Specific)	Layer 2 — Unified (NLP Pipeline Input)
Each source preserves its own natural schema. Raw Tweet Parquet — keeps engagement metrics Raw Reddit Parquet — keeps subreddit, upvote_ratio Raw News Parquet — keeps headline, source_name Source-specific fields are NOT discarded — they become candidates for feature engineering (e.g., tweet engagement as sentiment weight multiplier).	All sources merged into one clean Parquet file. Only 6 universal fields: doc_id, published_at, source, text, ticker, url This is the ONLY file fed into the NLP preprocessing pipeline. Engagement metrics re-enter the project during Feature Engineering as weighted sentiment aggregators.

Raw Tweet Parquet ■■■ Raw Reddit Parquet ■■■■■■ merge() ■■■ Unified Parquet ■■■
NLP Pipeline Raw News Parquet ■■■

8. Layer 1 — Tweets Schema

tweet_id | published_at | author | author_followers | text | likes | retweets |
replies | is_retweet | original_tweet_id | ticker | source

tweet_id

Unique identifier. Essential for deduplication — Kaggle datasets often contain duplicates.

published_at

MOST CRITICAL FIELD. Must be timezone-aware (UTC) because TSLA trades on NYSE (EST). Timezone misalignment silently corrupts trading signal labels.

author + author_followers

For TSLA, Elon Musk's tweets are categorically different from random users. author_followers becomes a credibility weight in feature engineering.

text

Raw unprocessed tweet. Never overwrite in Layer 1. Always preserve the original.

likes + retweets + replies

Engagement signals. A tweet with 500K retweets carries more market impact than one with 3. These three form the tweet influence score in Feature Engineering.

is_retweet + original_tweet_id

Critical for avoiding double-counting. 10,000 retweets of Elon's post must not be treated as 10,001 independent sentiment data points.

ticker

Always TSLA here, but kept for schema consistency with the unified layer.

source

Always 'twitter' here, needed for the merge step.

9. Layer 1 — Reddit Schema

```
post_id | published_at | subreddit | author | title | text | upvotes | upvote_ratio |  
num_comments | is_comment | parent_post_id | ticker | source
```

subreddit

VERY IMPORTANT. r/wallstreetbets (speculative/aggressive), r/investing (analytical), r/teslamotors (product-focused, not financial). These three communities have fundamentally different sentiment dynamics. Treating them identically would be academically weak.

title + text (separate columns)

Reddit posts have both a title and body. Many posts are title-only. Keep separate in Layer 1 — concatenation decision belongs in the NLP preprocessing step.

upvotes + upvote_ratio

Together more informative than upvotes alone. High upvotes + 0.95 ratio = community-validated. High upvotes + 0.51 ratio = controversial. Ratio is the signal quality indicator.

num_comments

High comment count correlates with community engagement and market impact.

is_comment + parent_post_id

Distinguishes top-level posts from comments. Needed to avoid treating a comment as standalone context when it is a reply to a specific post.

10. Layer 1 — News Schema

```
article_id | published_at | source_name | author | headline | content | url | ticker  
| source
```

article_id

Generated hash (e.g., MD5 of URL) since news APIs do not always provide a native ID.

published_at

Timezone awareness critical. European outlets may report after NYSE market close — this affects which trading day the article belongs to for label alignment.

source_name

Reuters vs. a small blog are not equal. Enables source credibility weighting in feature engineering, analogous to author_followers in tweets.

headline + content (separate columns)

Kept separate deliberately. Headline-only models vs. full-content models is a standard comparison in NLP-finance research. Separating them enables this experiment without re-collecting data.

url

Primary deduplication key for news. The same article often appears across multiple aggregators with slightly different text.

11. Layer 2 — Unified Schema

```
doc_id | published_at | source | text | ticker | url
```

doc_id

New generated ID for the unified dataset. IDs from different sources have different formats and would clash if merged directly.

published_at

Carried over, now guaranteed UTC-normalized across ALL sources.

source

Single most important field for downstream analysis. Values: twitter, reddit_post, reddit_comment, news. Enables filtering and source-specific comparison at any point.

text

Tweets: raw tweet text. Reddit: title + ' ' + body (with null handling). News: headline + ' ' + content. This concatenation is the ONLY transformation between layers.

ticker

Always TSLA here, but kept for schema reusability if the project ever expands.

url

Final deduplication check across sources — the same story can appear in news and get linked in a Reddit post.

What gets dropped between layers: All engagement metrics (likes, retweets, upvotes, num_comments, author_followers) are intentionally excluded from Layer 2. They are NOT lost — they remain in Layer 1 Parquet files and re-enter the project during Feature Engineering as weighted sentiment aggregators. Keeping them in the unified NLP layer would add noise to preprocessing.

12. Key Architectural Decisions Summary

Decision	Justification
Single sector + single stock	Deeper academic analysis, cleaner modeling, easier to defend before jury.
Tesla (TSLA)	Musk Effect documented in literature, strong Reddit/news presence, clean sentiment dynamics.
Time window 2020–2023	Four distinct market regimes → more robust and generalizable model.
Pre-collected tweet datasets	Twitter API closed in 2023. Kaggle/HuggingFace datasets are the only free option for historical tweets.
Two-layer schema	Preserves source-specific fields (critical for feature engineering) while providing a clean unified input to the NLP pipeline.
UTC timezone normalization	Silent killer of label accuracy if ignored. Must be enforced at collection.
title/headline + body separate	Enables headline-only vs. full-content model comparison in thesis.
Engagement metrics in Layer 1 only	Avoids NLP pipeline noise while preserving weighting capability for feature engineering phase.
Trading signal labels (BUY/SELL/HOLD)	Most ambitious and academically rich target variable. Requires explicit threshold definition and class imbalance mitigation strategy.

This document summarizes the data collection brainstorming session for the PFE project 'Sentiment Analysis for Financial Market Prediction'. Generated: March 12, 2026. Next step: implement the collection pipeline for each source following the schema architecture above.